



Gen AI Prompt Engineering

By 

Components of prompt

1. Role (Who the AI should be?)

Defines the persona, expertise, or perspective the AI should assume while responding. This helps control tone, depth, and style of the output.

Example:

“You are a subject-matter expert in artificial intelligence explaining concepts to beginners.”

2. Task (What needs to be done?)

The overall objective or purpose of the prompt. Clearly states what the AI is expected to accomplish.

Example:

“Generate a summary of an article.”

3. Instruction (How should the task be performed?)

Specifies how the task should be executed, including rules, limits, or style.

Example:

“Summarize the article in no more than 100 words, keeping the key points intact.”

Role

Who is AI

Task

Action or process

Instruction

Rules to process

Input

Actual content,
dataset, image

context

Background Info

Output Format

How response look

Components of prompt

4. Input (What data is provided to perform the task?)

The actual content or information on which the AI operates.

Example:

“The article text:

‘AI is transforming industries by automating tasks...’”

5. Context (Additional details that help improve accuracy)

Background information, constraints, target audience, or usage scenario that refines the response.

Example:

“This summary is intended for a general audience with no technical background.”

6. Output Format (How should the response look?)

Defines the structure, layout, or presentation style of the final output.

Example:

“Present the summary as a single paragraph using simple language.”

“Get the change for 1000 in 100 rupees notes”

Role

Who is AI

Task

Action or process

Instruction

Rules to process

Input

Actual content,
dataset, image

context

Background Info

Output Format

How response look

Components of prompt

Task

Action or process

Explain how neural networks learn.

Role

Who is AI

Explain how neural networks learn. **You are an AI expert explaining concepts to complete beginners.**

Instruction

Rules to process

Explain how neural networks learn. You are an AI expert explaining concepts to complete beginners. **Keep the explanation simple, avoid jargon, and use everyday language.**

Input

Actual content, dataset,
image

Explain how neural networks learn. You are an AI expert explaining concepts to complete beginners. Keep the explanation simple, avoid jargon, and use everyday language. **A neural network takes in data through an input layer, processes it through hidden layers, and produces a result through an output layer.**

context

Background Info

Explain how neural networks learn. You are an AI expert explaining concepts to complete beginners. Keep the explanation simple, avoid jargon, and use everyday language. A neural network takes in data through an input layer, processes it through hidden layers, and produces a result through an output layer. **This is for first-year engineering students in their very first week of an AI course.**

Output Format

How response look

Explain how neural networks learn. You are an AI expert explaining concepts to complete beginners. Keep the explanation simple, avoid jargon, and use everyday language. Here is the input text to work from: "A neural network takes in data through an input layer, processes it through hidden layers, and produces a result through an output layer." This is for first-year engineering students in their very first week of an AI course. **Present the answer in a JSON format**

Prompt Example 1

Role:

You are an experienced computer science trainer who explains technical topics in a simple way.

Task:

Explain the given concept clearly.

Instruction:

1. Start with a simple definition.
2. Explain the concept step-by-step.
3. Provide a real-world analogy.
4. Give a short example.

Input:

Concept: Neural Networks

Context:

The explanation is for beginners who have no background in artificial intelligence.

Output Format:

Definition:

<simple definition>

Explanation:

<step-by-step explanation>

Analogy:

<real-world analogy>

Example:

<short example>

Prompt Example 2

Role:

You are a professional data analyst specializing in retail analytics.

Task:

Analyze the sales data and identify the best and worst performing products.

Instruction:

1. Determine the product with the highest revenue.
2. Determine the product with the lowest revenue.
3. Provide one insight about the sales trend.

Input:

Product A – Revenue: \$5000

Product B – Revenue: \$2000

Product C – Revenue: \$7000

Product D – Revenue: \$1500

Context:

The analysis will be shown to a retail store manager who wants quick insights to improve product sales.

Output Format:

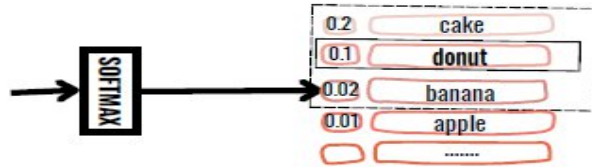
Top Product: <product name> – <revenue>

Lowest Product: <product name> – <revenue>

Key Insight: <short explanation>

Prompt parameters

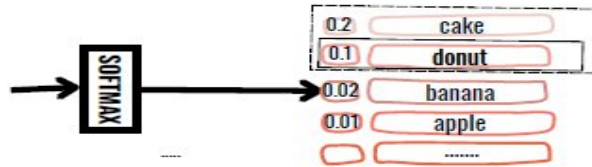
Top N.



Top N : The word/token is selected using random-weighted strategy but only from amongst the Top 'N' words/tokens

Here for $N=3$, one of cake, donut or banana will be selected randomly but apple will never be selected

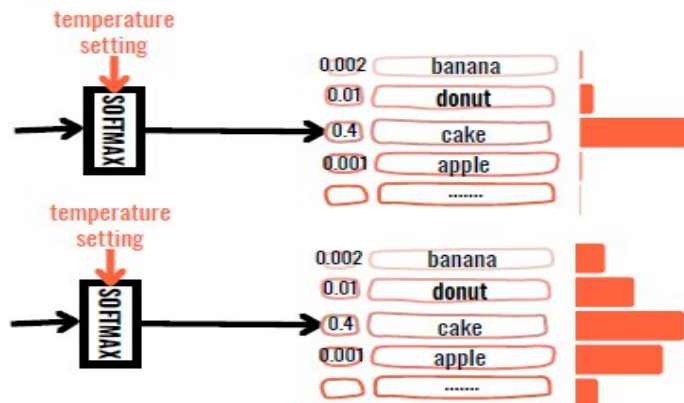
Top P.



Top P : The word/token is selected using random-weighted strategy but only from amongst the top words totalling to probability $\leq P$

Here for $P=0.33$, one of cake or donut will be selected randomly but apple or banana will never be selected

Temperature



Cooler Temperature (lesser value) :
The distribution is strongly peaked

Warmer Temperature (higher value) :
Flatter probability distribution

Terminology

Term	Meaning
Prompt	Instructions + context; what you send to the model
Generated output	Tokens the model produces
Context window / Max sequence length	Total tokens model can handle at once = prompt + output
max_tokens / max_output_tokens	Limits how long the model output can be; must fit inside context window
Sampling Parameters	temperature, top-p, max_output_tokens

Prompt Engineering

Prompt Engineering is the art of designing inputs for LLMs (like ChatGPT) to obtain desired responses.

Prompt Engineering Techniques :

1. Zero-Shot Prompting
2. Few-Shot Prompting
3. Chain-of-Thought (CoT) Prompting
4. ReAct (Reasoning + Acting)
5. Self-Consistency
6. Instruction-Based Prompting
7. Role-Based Prompting

Zero Shot Prompting

Concept

Zero-shot prompting relies on the model's pre-trained knowledge to answer queries without providing examples.

Example

Prompt:

"What is the capital of France?"

ChatGPT Response:

"The capital of France is Paris."

Use Case:

Simple fact-based queries

General knowledge retrieval

Limitations:

Struggles with tasks requiring context or reasoning

Few Shot Prompting

Concept

Providing a few examples to guide the model on the desired format and reasoning.

Example

Prompt:

"Classify the following texts as Positive or Negative sentiment:

'I love this product!' → Positive

'This is the worst experience ever!' → Negative

'I am very happy with my purchase.' → ?"*

ChatGPT Response:

"Positive."

Use Case:

Sentiment analysis

Text classification

Pattern learning

Limitations:

Requires carefully selected examples

Chain of Thought (CoT)

Chain-of-Thought (CoT) Prompting

Concept

Encouraging the model to break down reasoning into intermediate steps before answering.

Example

Prompt:

"A farmer has 3 baskets. Each basket contains 5 apples. If he gives away 4 apples, how many apples remain? Let's think step by step."

ChatGPT Response:

"Step 1: The farmer starts with 3 baskets $\times$ 5 apples = 15 apples.

Step 2: He gives away 4 apples.

Step 3: $15 - 4 = 11$ apples remain.

Final Answer: 11 apples."

Use Case:

Complex math problems

Logical reasoning tasks

Limitations:

Requires structured reasoning tasks

ReAct

ReAct (Reasoning + Acting) Reason → Action → Observation → Response

Concept

Combining reasoning with action by making the model think, retrieve, and act iteratively.

Example

Prompt:

"Who won the 2023 FIFA World Cup? Think step by step and search if needed."

ChatGPT Response (simulating ReAct):

"Step 1: The last FIFA World Cup was held in 2022, and Argentina won.

Step 2: Let me check the 2023 winner (simulated search).

Step 3: The 2023 FIFA Women's World Cup was won by Spain.

Final Answer: Spain won the 2023 FIFA Women's World Cup."

Use Case:

Dynamic decision-making

Research-based reasoning

Limitations:

Needs external search for live data

Self-Consistency

Run the same prompt multiple times

Example

Prompt:

"If a train travels 100 miles in 2 hours, what is its speed?

or

"If a train travels 100 miles in 2 hours, what is its speed? Solve using two different methods." <not exactly a self consistency)

ChatGPT Response:

Method 1:

"Speed = Distance ÷ Time = 100 miles ÷ 2 hours = 50 mph."

Method 2:

"If the train covers 100 miles in 2 hours, in 1 hour it covers 50 miles. So, the speed is 50 mph."

"Final Answer: 50 mph."

Use Case:

Reducing hallucinations in reasoning tasks

Improving mathematical accuracy

Limitations:

Can be computationally expensive

Other useful Techniques

Instruction-Based Prompting

Instead of a vague query, clearly instruct the model.

Example

✗ "Summarize this article."

✓ "Summarize this article in three bullet points, each under 20 words."

Role-Based Prompting

Assigning a persona to the model to align responses.

Example

"You are a cybersecurity expert. Explain the importance of multi-factor authentication in banking security."

By using these techniques, you can significantly improve LLM's responses for different use cases like NLP tasks, logical reasoning, and automation



Thank you